

Savitzky-Golay optimizer diagnostics

Objective: mean squared translational/angular acceleration residual; angular residual uses the reference inertia/mass metric

Tolerances: {'ftol': None, 'xtol': None, 'gtol': 1e-06, 'note': 'ftol and xtol are disabled by default in the SG estimator; gtol or max_nfev terminates the solve unless explicitly overridden'}

Selected strict-ZOH solve

status=1 success=True nfev=4 njev=4

message='gtol' termination condition is satisfied.

reported optimality=8.29991506634542e-08

initial={'cost': 1.506118527187727, 'gradient_inf_norm': 0.05407113691693155, 'gradient_l2_norm': 0.08131918613579661, 'jacobian': {'condition_number_nonzero_subspace': 390.5056559014383, 'numerical_rank': 14, 'numerical_tolerance': 7.626041346959729e-11, 'singular_values': [119.169455130048, 118.05129158680154, 102.5225129673429, 48.59877297969992, 44.72324423924553, 25.862391786987917, 12.9862922783641, 11.726552611821571, 11.309322790960085, 11.038041064786228, 9.885735105655161, 3.7962499907097054, 3.7934371275992755, 0.30516704003930173]}, 'residual_l2_norm': 1.7355797458991777}

final={'cost': 1.5061181148746856, 'gradient_inf_norm': 8.29991506634542e-08, 'gradient_l2_norm': 1.3402711405330143e-07, 'jacobian': {'condition_number_nonzero_subspace': 390.70662516062816, 'numerical_rank': 14, 'numerical_tolerance': 7.62601507070362e-11, 'singular_values': [119.16904452027276, 118.05126216589167, 102.52250673106315, 48.59885446704944, 44.72326120794145, 25.86241352978537, 12.9862709757674, 11.726433534242567, 11.309258868738604, 11.037874315716792, 9.885604135158369, 3.7962642557868858, 3.793425991523321, 0.30500901916183204]}, 'residual_l2_norm': 1.7355795083341388}

coordinate_step_l2_norm=0.00026337342771118235

cost_reduction=4.1231304148681147e-07

gradient_tolerance_satisfied=True

finite_difference_max_relative_error=7.599270019489056e-12

finite_difference_directional_checks=[{'analytic_l2_norm': 98.35043287577011, 'difference_l2_norm': 7.473914959591853e-10, 'direction': 'negative_gradient', 'finite_difference_l2_norm': 98.35043287610891, 'relative_error': 7.599270019489056e-12, 'scheme': 'central', 'step': 6.0554544523933395e-06}]

All smoothstep and strict-ZOH stage details are in optimizer_diagnostics.json.